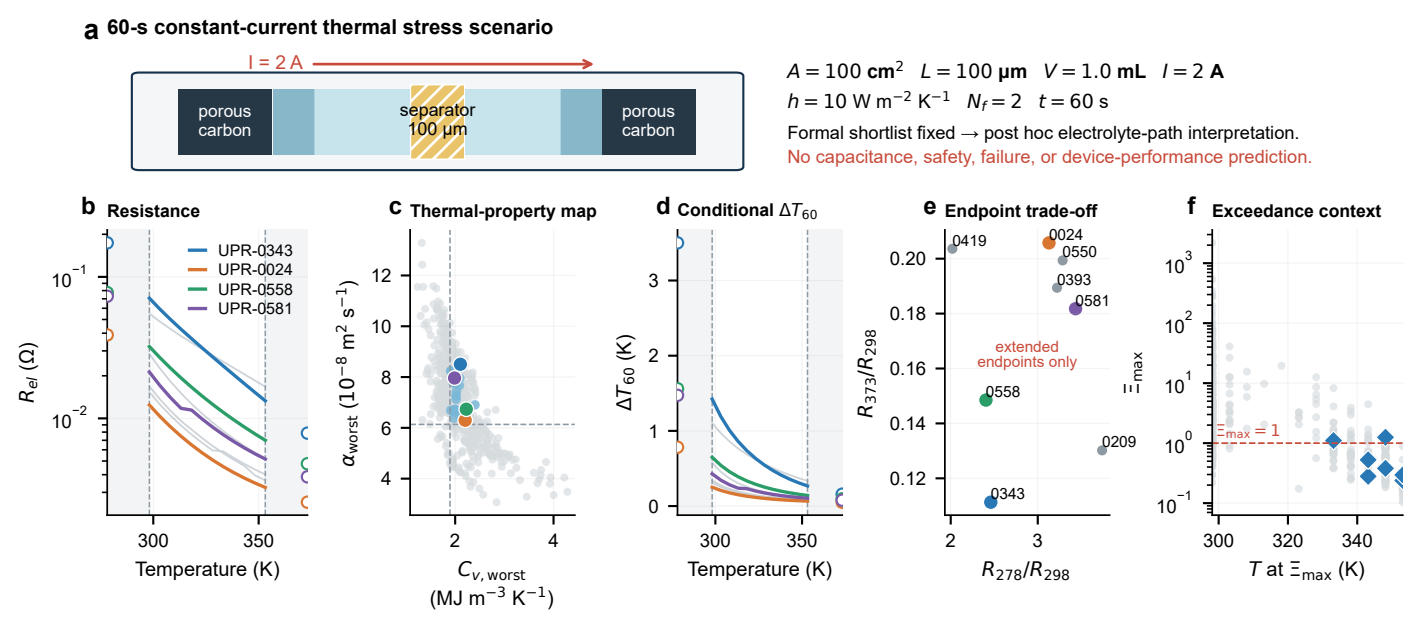

a 60-s constant-current thermal stress scenario
I = 2 A
porous carbon
separator 100 µm
porous carbon
A = 100 cm² L = 100 µm V = 1.0 mL I = 2 A
h = 10 W m⁻² K⁻¹ N_f = 2 t = 60 s
Formal shortlist fixed → post hoc electrolyte-path interpretation.
No capacitance, safety, failure, or device-performance prediction.
b Resistance
UPR-0343
UPR-0024
UPR-0558
UPR-0581
R_el (Ω)
Temperature (K)
c Thermal-property map
α_worst (10⁻⁸ m² s⁻¹)
C_v, worst (MJ m⁻³ K⁻¹)
d Conditional ΔT60
ΔT60 (K)
Temperature (K)
e Endpoint trade-off
0419
0024
0550
0393
0581
extended endpoints only
0558
0209
0343
R373/R298
R278/R298
f Exceedance context
Ξmax
Ξmax = 1
T at Ξmax (K)